# Requirements Specification and Validation

## Lecture 5

# Agenda

- Requirements Specification Templates
- Characteristics of Quality Requirements
- Guidelines to Specify Requirements
- Requirements Validation
- Validation Criteria
- Verification and Validation

# Requirements Specification Templates

INFO



## *Software Requirements Specification Template*

A *software requirements specification* (SRS) is a work product that is created when a detailed description of all aspects of the software to be built must be specified before the project is to commence. It is important to note that a formal SRS is not always written. In fact, there are many instances in which effort expended on an SRS might be better spent in other software engineering activities. However, when software is to be developed by a third party, when a lack of specification would create severe business issues, or when a system is extremely complex or business critical, an SRS may be justified.

Karl Wiegers [Wie03] of Process Impact Inc. has developed a worthwhile template (available at **www.processimpact.com/process_assets/srs_template.doc**) that can serve as a guideline for those who must create a complete SRS. A topic outline follows:

**Table of Contents**

**Revision History**

1. **Introduction**
   - 1.1 Purpose
   - 1.2 Document Conventions
   - 1.3 Intended Audience and Reading Suggestions
   - 1.4 Project Scope
   - 1.5 References
2. **Overall Description**
   - 2.1 Product Perspective
   - 2.2 Product Features
   - 2.3 User Classes and Characteristics
   - 2.4 Operating Environment
   - 2.5 Design and Implementation Constraints
   - 2.6 User Documentation
   - 2.7 Assumptions and Dependencies
3. **System Features**
   - 3.1 System Feature 1
   - 3.2 System Feature 2 (and so on)
4. **External Interface Requirements**
   - 4.1 User Interfaces
   - 4.2 Hardware Interfaces
   - 4.3 Software Interfaces
   - 4.4 Communications Interfaces
5. **Other Nonfunctional Requirements**
   - 5.1 Performance Requirements
   - 5.2 Safety Requirements
   - 5.3 Security Requirements
   - 5.4 Software Quality Attributes
6. **Other Requirements**

**Appendix A: Glossary**

**Appendix B: Analysis Models**

**Appendix C: Issues List**

A detailed description of each SRS topic can be obtained by downloading the SRS template at the URL noted in this sidebar.

Pressman and Maxim

# Contents


## Project Drivers

1. The Purpose of the Project
2. The Stakeholders

## Project Constraints

3. Mandated Constraints
4. Naming Conventions and Terminology
5. Relevant Facts and Assumptions

## Functional Requirements

6. The Scope of the Work
7. Business Data Model and Data Dictionary
8. The Scope of the Product
9. Functional Requirements

## Non-functional Requirements

10. Look and Feel Requirements
11. Usability and Humanity Requirements
12. Performance Requirements
13. Operational and Environmental Requirements
14. Maintainability and Support Requirements
15. Security Requirements
16. Cultural Requirements
17. Legal Requirements

## Project Issues

18. Open Issues
19. Off-the-Shelf Solutions
20. New Problems
21. Tasks
22. Migration to the New Product
23. Risks
24. Costs
25. User Documentation and Training
26. Waiting Room
27. Ideas for Solutions


Robertson and Robertson

**Table of Contents**

1. Introduction
   1.1 Purpose
   1.2 Scope
   1.3 Definitions, acronyms, and abbreviations
   1.4 References
   1.5 Overview
2. Overall description
   2.1 Product perspective
   2.2 Product functions
   2.3 User characteristics
   2.4 Constraints
   2.5 Assumptions and dependencies
3. Specific requirements (See 5.3.1 through 5.3.8 for explanations of possible specific requirements. See also Annex A for several different ways of organizing this section of the SRS.)

Appendixes

Index

IEEE Std 830-1998

1. **Introduction**
    - 1.1 System purpose
    - 1.2 System scope
    - 1.3 System overview
        - 1.3.1 System context
        - 1.3.2 System functions
        - 1.3.3 User characteristics
    - 1.4 Definitions
2. **References**
3. **System requirements**
    - 3.1 Functional requirements
    - 3.2 Usability requirements
    - 3.3 Performance requirements
    - 3.4 System interface
    - 3.5 System operations
    - 3.6 System modes and states
    - 3.7 Physical characteristics
    - 3.8 Environmental conditions
    - 3.9 System security
    - 3.10 Information management
    - 3.11 Policies and regulations
    - 3.12 System life cycle sustainment
    - 3.13Packaging, handling, shipping and transportation
4. **Verification**
    - (parallel to subsections in Section 3)
5. **Appendices**
    - Assumptions and dependencies
    - Acronyms and abbreviations

ISO/IEC/IEEE 29148

# Características de Requerimientos de Calidad

# Características de Requerimientos de Calidad

1. Correctos
2. No ambiguos
3. Entendibles
4. Completos
5. Consistentes
6. Factibles


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# Características de Requerimientos de Calidad

7. Priorizados
8. Verificables
9. Modificable
10. Trazables
11. Organizados
12. Independientes del diseño


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# Características de Requerimientos de Calidad

- **Correcto**
  - Cada requerimiento en el documento representa algo que el sistema debe cumplir.
- **No ambiguo**
  - Si y solo puede estar sujeto a una interpretación.
- **Entendible**
  - Debe utilizar el lenguaje y la cultura del stakeholder y presentar los requerimientos de una forma entendible a los stakeholders y a los desarrolladores

# Características de Requerimientos de Calidad

- **Completo**
  - Un documento de especificación de requerimientos está completo si y solo si describe todos los requerimientos de usuario y los requerimientos detallados asociados con la función, desempeño y restricciones.
- **Consistente**
  - Si ningún requerimiento entra en conflicto con algún otro.

# Características de Requerimientos de Calidad

- **Factible**
  - Evaluar si es técnicamente posible su implementación
- **Priorizado**
  - Utilizando atributos del requerimiento se debe determinar si debe ser incluido al final de producto y el orden de inclusión en un proceso de desarrollo iterativo.
  - MOSCOW
    - ❖Debe ser entregado en esta iteración (Musts)
    - ❖Se espera que sea entregado (Shoulds)
    - ❖Sería bueno que se implemente (Coulds)
    - ❖Definitivamente no (Won’ts)

# Características de Requerimientos de Calidad

- **Verificable**
  - Un documento de especificación de requerimientos es verificable si y solo si cada uno de los requerimientos lo es.
  - El determinar que un software cumple con la especificación debe ser costo-efectivo
- **Modificable**
  - Si los requerimientos pueden ser cambiados y el documento permanece consistente y completo.

# Características de Requerimientos de Calidad

- **Trazable**
  - Un documento de requerimientos es trazable si y solo si existe un mecanismo para rastrear cada requerimiento a través de todo el desarrollo.
- **Organizado**
  - El documento de especificación de requerimientos debe ser organizado, por ejemplo, con base en estándares, una lista de contenidos, un índice, incluir un glosario de términos y un historial de revisiones. Además, deben ser categorizados en grupos de requerimientos.


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# Características de Requerimientos de Calidad

- **Independientes del diseño**
  - Si ningún requerimiento en el documento de especificación de requerimientos especifica arquitectura de software o algoritmos.

# Características de Requerimientos de Calidad

- **Recomendaciones generales**:
  - Eliminar términos imprecisos tales como: algunos, algunas veces, usualmente, la mayoría, a menudo, frecuentemente.
  - Eliminar etc., tales como.
  - ***Los rangos deben ser explícitos.***
  - Cuando se especifica un cálculo, se deben dar al menos dos ejemplos.
  - Eliminar pronombres o sustantivos ambiguos. Por ejemplo: bandera.


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# Lineamientos para Elaborar Requerimientos

# Lineamientos para Elaborar Requerimientos

- Los requerimientos de usuarios generalmente implican verbos tales como proveer, determinar, mejorar, reducir.
- Los requerimientos del sistema implican acciones tales como aceptar, chequear, seleccionar, calcular.
- Formalmente defina todos los términos.
- Use la terminología del dominio del problema.


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# Lineamientos para Elaborar Requerimientos

- Nunca use el mismo término para diferentes cosas.
- Siempre use el mismo término para referirse a la misma cosa.
- Escriba para el lector (revise que su requerimiento sea fácil de entender, no ambiguo, etc.)

# Lineamientos para Elaborar Requerimientos

- Use listas y numeración de referencia
- Sea flexible (no tiene que utilizar un formato o técnica en particular)
- Organice los requerimientos apropiadamente. Es decir, usuario, hardware, interfaces de software.

# Lineamientos para Elaborar Requerimientos

- Provea índice de contenido, referencias, referencias cruzadas
- Trate de no duplicar requerimientos, defina el requerimiento general y luego los específicos

# Lineamientos para Elaborar Requerimientos

- Evite generalidades como “todos los requerimientos deben ser probados”.
- Referencie otros documentos claramente (no copie).
- Use layout y fuentes consistentemente.
- Lea lo que usted escribe.

# Validación de Requerimientos

# Validación de Requerimientos

- Requirements validation is the process of checking that requirements define the system that the customer really wants.
- It overlaps with elicitation and analysis, as it is concerned with finding problems with the requirements.
- Requirements validation is critically important because errors in a requirements document can lead to extensive rework costs when these problems are discovered during development or after the system is in service.

# Validación de Requerimientos

- During the requirements validation process, different types of checks should be carried out on the requirements in the requirements document such as:
  - Validity checks: These check that the requirements reflect the real needs of system users.
  - Consistency checks: Requirements in the document should not conflict. That is, there should not be contradictory constraints or different descriptions of the same system function.
  - Completeness checks: The requirements document should include requirements that define all functions and the constraints intended by the system user.
  - Realism checks: By using knowledge of existing technologies, the requirements should be checked to ensure that they can be implemented within the proposed budget and schedule for the system.
  - Verifiability: To reduce the potential for dispute between customer and contractor, requirements should always be written so that they are verifiable.
    - ❖ This means writing a set of tests that can demonstrate that the system meets the specified requirement.

# Validación de Requerimientos

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## *Requirements Validation Checklist*

It is often useful to examine each requirement against a set of checklist questions. Here is a small subset of those that might be asked:

- Are requirements stated clearly? Can they be misinterpreted?
- Is the source (e.g., a person, a regulation, a document) of the requirement identified? Has the final statement of the requirement been examined by or against the original source?
- Is the requirement bounded in quantitative terms?
- What other requirements relate to this requirement? Are they clearly noted via a cross-reference matrix or other mechanism?
- Does the requirement violate any system domain constraints?
- Is the requirement testable? If so, can we specify tests (sometimes called validation criteria) to exercise the requirement?
- Is the requirement traceable to any system model that has been created?
- Is the requirement traceable to overall system/product objectives?
- Is the specification structured in a way that leads to easy understanding, easy reference, and easy translation into more technical work products?
- Has an index for the specification been created?
- Have requirements associated with performance, behavior, and operational characteristics been clearly stated? What requirements appear to be implicit?

# Requirements Validation Techniques

- Requirements validation techniques can be used individually or in conjunction with one another:
  - Requirements reviews: The requirements are analysed systematically by a team of reviewers who check for errors and inconsistencies.
  - Prototyping: This involves developing an executable model of a system and using this with end-users and customers to see if it meets their needs and expectations.
  - Test-case generation: Requirements should be testable. If the tests for the requirements are devised as part of the validation process, this often reveals requirements problems.
    - ❖If a test is difficult or impossible to design, this usually means that the requirements will be difficult to implement and should be reconsidered.

# Criterios de Verificabilidad de Requerimientos

# Criterios de Verificabilidad de Requerimientos

- También conocidos como criterios de validación (validation criteria).
- Definen cómo el cliente chequeará que los requerimientos han sido satisfechos.
  1. Aprobación de un experto.
  2. Una prueba medible
     a. por un periodo definido de tiempo, o
     b. con un número definido de usuarios, o
     c. con un número de casos de prueba.
  3. Pruebas de aceptación de usuario.
  4. Simulación.
  5. Encuestas de satisfacción a usuarios.

# Criterios de Verificabilidad de Requerimientos

- El sistema debe ser implementado en tres módulos principales, uno para cada función: detección, registro y análisis estadístico.
  - Organisational (development)
  - Criterio:
  - El Director de Calidad debe confirmar que el sistema ha sido diseñado en tres módulos principales, uno para cada función: detección, registro y análisis estadístico.


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# Criterios de Verificabilidad de Requerimientos

- El tamaño máximo del archivo de entrada debe ser dos millones de caracteres y los archivos de este tamaño deben ser convertidos en menos de 60 segundos.
  - Product efficiency (space) y efficiency (performance)
  - Criterio:
  - Durante las pruebas Beta ningún archivo de menos de dos millones de caracteres tomará más de 60 segundos en ser convertido.


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# Criterios de Verificabilidad de Requerimientos

- ECUSystem estará fuera de operación para servicio no más de 0.001% de su año operativo.
  - Product (dependability)
  - Criterio:
  - La operación será simulada por un periodo de 100 horas y ECUSystem estará disponible el 99.999% del tiempo.

# Criterios de Verificabilidad de Requerimientos

- Información no relevante del paciente será eliminada luego de 6 meses de terminado el tratamiento.
  - External (ethical/regulatory)
  - Criterio:
  - El sistema será probado con detalles de 100 pacientes, las pruebas simularán el escenario de 6 meses luego de terminado el tratamiento y el cliente confirma que los detalles del paciente son borrados. El Director Jurídico confirma que el sistema cumple con el Código Profesional de Ética.

# Verification and Validation (V&V)

# Verification and Validation

- Verification and validation are not the same thing, although they are often confused.
- Software **verification** is the **process** of checking that the software **meets** its stated functional and non-functional **requirements**.
- The aim of software **validation** is to **ensure** that the software meets the **customer's expectations**. It goes beyond checking conformance with the specification to demonstrating that the software does what the customer expects it to do.


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# Verification and Validation

- **Validation:** *Are we building the right product?*
- **Verification:** *Are we building the product right?*

- Activities that play an extremely important role in V&V include:
  - Software testing
  - Software inspections and reviews
  - Quality and configuration audits
  - Performance monitoring
  - Simulation
  - Algorithm analysis
  - Documentation review.

# Next On

# Take Away Points

- What are the characteristics of quality requirement?
- Suitability of each validation criteria for a given requirement.
- Identify the main elements of a requirements specification document.
- Guidelines to Specify Requirements
- What is the difference between validation and verification?

# Further Reading

- Robertson S. and Robertson J, Mastering the Requirements Process: Getting Requirements Right
- K. Wiegers and J. Beatty, “Software Requirements (Developer Best Practices) 3rd edition”
- Systems and software engineering - Life cycle processes – Requirements engineering
- The IEEE Guide to the Software Engineering Body of Knowledge – SWEBOK
  - Chapters 3 to 6

# Further Reading

- Pressman and Maxim, “Software Engineering”
  - Chapter 8: Understanding Requirements
  - Chapters 9 to 11
- Ian Somerville, “Software Engineering”
  - Chapter 4: Requirements Engineering
- IEEE Recommended Practice for Software Requirements Specifications
- Volere, Volere Requirements Resources
  - https://www.volere.org/templates/

# Next Lecture

- Management of Software Projects